TMI-ToE 1.0
Interface-Field Architecture for the Unification of Fundamental Interactions

Theory of Manifold Interfaces

Version 1.0

Contents

Abstract

This document fixes the module **TMI-ToE 1.0** within the Theory of Manifold Interfaces. Its purpose is not to claim a completed "Theory of Everything", but to define a minimal consistent architecture in which the four fundamental interactions are described as projections of a unified interface curvature.

Short status:

$$\boxed{\text{TMI-ToE 1.0} \neq \text{a complete Theory of Everything}}$$

$$\boxed{\text{TMI-ToE 1.0 is a minimal interface-field architecture of unification.}}$$

Main idea:

$$\boxed{F_a = \pi_a(\mathbb{F}_I), \qquad a \in \{G, S, W, EM\}}$$

where $\mathbb{F}_I$ is the universal interface curvature, and F_G, F_S, F_W, F_{EM} are the gravitational, strong, weak, and electromagnetic regimes.

1 Initial status of the module

The physical branch of TMI already contains several modules:

- **TMI-QG**: interface geometric decoherence, the quantity Γ_I, a testable hypothesis, and a statistical form.
- **TMI-Dark**: the dark sector as a regime of the interface field $\chi_I = \chi_0 + \delta\chi$.
- **TMI-BH**: a black hole as a one-way causal-informational interface.
- **TMI-ToE**: an architecture unifying these physical layers through a universal connection, curvature, and interface field.

TMI-ToE does not replace established theories. It requires general relativity, the Standard Model, and quantum mechanics to be recovered as limiting cases.

2 TMI-ToE-0.9: consistency conditions

The goal of version 0.9 is not to add new concepts, but to close the technical conditions of admissibility. If any of these conditions fails, the current TMI-ToE version must be revised.

2.1 Recovery of known theories

General-relativistic limit

If the interface sector is switched off:

$$\chi_I \to 0, \qquad \mathcal{L}_{mix} \to 0,$$

then:

$$\boxed{S_{TMI-ToE} \to S_{GR} + S_{SM}}.$$

If matter and gauge fields are also removed:

$$\mathcal{L}_{SM} \to 0,$$

then:

$$\boxed{S_{TMI-ToE} \to S_{GR}}.$$

Standard Model limit

For a fixed geometry:

$$g_{\mu\nu} = g_{\mu\nu}^{fixed}$$

and vanishing interface mixing:

$$\chi_I \to 0, \qquad \zeta_a, \eta_H, y_I \to 0,$$

one must obtain:

$$\boxed{S_{TMI-ToE} \to S_{SM}}.$$

Thus the low-energy gauge limit must contain

$$\boxed{SU(3) \times SU(2) \times U(1)}.$$

Quantum limit

For a fixed background and a weak interface sector, standard quantum dynamics must be recovered:

$$\boxed{i\hbar \frac{\partial}{\partial t} \Psi = \widehat{H}\Psi}$$

or the corresponding quantum field-theoretic form on a fixed background.

2.2 The universal group G_I

The minimal requirement is:

$$\boxed{G_I \supset \mathrm{Spin}(1,3) \times SU(3) \times SU(2) \times U(1)}.$$

For version 1.0, the safest form is not to assert a new simple group, but to fix an inclusion structure:

$$\boxed{G_I = G_{grav} \ltimes G_{SM} \ltimes G_\chi}$$

where

$$G_{grav} \supset \mathrm{Spin}(1,3), \qquad G_{SM} = SU(3) \times SU(2) \times U(1).$$

Thus G_I is the group of admissible interface transformations containing gravitational, gauge, and interface symmetries.

2.3 Projection consistency

We introduce algebra projections:

$$\boxed{\pi_a : \mathfrak{g}_I \to \mathfrak{g}_a, \qquad a \in \{G, S, W, EM\}},$$

where $\mathfrak{g}_I = \mathrm{Lie}(G_I)$.

Then the physical curvatures are:

$$\boxed{F_G = \pi_G(\mathbb{F}_I)}, \qquad \boxed{F_S = \pi_S(\mathbb{F}_I)},$$

$$\boxed{F_W = \pi_W(\mathbb{F}_I)}, \qquad \boxed{F_{EM} = \pi_{EM}(\mathbb{F}_I)}.$$

2.4 Gauge invariance

The mixing term must preserve the gauge structure of the Standard Model. A safe form is:

$$\boxed{\mathcal{L}_{mix}^{gauge} = -\frac{1}{4}\sum_{a=1}^{3} \zeta_a \frac{\chi_I^2}{M_I^2} \mathrm{Tr}(F_{a\mu\nu}F_a^{\mu\nu})}$$

which is admissible if χ_I is a gauge singlet:

$$\boxed{\chi_I \sim (1,1,0)}.$$

2.5 Fermionic consistency

A raw term $\chi_I \bar{\Psi}\Psi$ may violate gauge invariance before electroweak symmetry breaking. A safer coupling to fermions is:

$$\boxed{\mathcal{L}_{\chi\Psi} = \sum_f y_{I,f} \frac{\chi_I}{M_I} \bar{\Psi}_{L,f} H \Psi_{R,f} + h.c.}$$

where H is the Higgs field and f indexes fermion type and generation.

2.6 Higgs sector

A compatible potential is:

$$\boxed{V(H,\chi_I) = -\mu_H^2 H^\dagger H + \lambda_H (H^\dagger H)^2 + V(\chi_I) + \eta_H \chi_I^2 H^\dagger H}$$

where

$$\boxed{V(\chi_I) = V_0 + \frac{1}{2}m_I^2\chi_I^2 + \frac{\beta_I}{4}\chi_I^4}.$$

The effective Higgs parameter is:

$$\boxed{\mu_{H,eff}^2 = \mu_H^2 - \eta_H \chi_I^2}.$$

Thus the Higgs vacuum expectation value may receive an interface correction:

$$\boxed{v_H^{eff} = v_H(\chi_I)}.$$

2.7 Anomaly cancellation

If χ_I is a gauge singlet and no new chiral fermions are added, the anomaly structure of the Standard Model remains unchanged:

$$\boxed{\mathcal{A}_{TMI-ToE} = \mathcal{A}_{SM} = 0}.$$

If new fermions or new G_I representations are introduced in future versions, one must require:

$$\boxed{\mathcal{A}_{G_I} = 0}.$$

2.8 EFT status

The present version must be treated as an effective interface-field theory:

$$\boxed{TMI\text{-}ToE = \text{effective interface-field theory}}$$

valid at energies

$$\boxed{E \ll M_I}.$$

Operators such as

$$\frac{\chi_I^2}{M_I^2} F_{\mu\nu} F^{\mu\nu}, \qquad \frac{\chi_I}{M_I} \bar{\Psi}_L H \Psi_R$$

are effective operators suppressed by the interface scale M_I.

2.9 Stability

The potential must be bounded from below. Sufficient conditions are:

$$\boxed{\beta_I > 0, \qquad \lambda_H > 0}.$$

The mixed potential must not generate an unstable vacuum. A working condition is:

$$\boxed{\eta_H > -2\sqrt{\lambda_H \beta_I / 4}}.$$

2.10 Compatibility with other TMI modules

Connection with TMI-QG

$$\boxed{\chi_I \Rightarrow \Gamma_I}, \qquad \boxed{\Delta\Gamma_I = \kappa_I x_I}.$$

Connection with TMI-Dark

$$\boxed{\chi_I = \chi_0 + \delta\chi}, \qquad \boxed{\chi_0 \Rightarrow DE, \quad \delta\chi \Rightarrow DM}.$$

Connection with TMI-BH

On a black-hole horizon, the restrictions

$$\boxed{\mathbb{A}_I|_{\mathcal{H}_g}}, \qquad \boxed{\mathbb{F}_I|_{\mathcal{H}_g}}$$

must have a meaningful interpretation.

3 TMI-ToE-1.0: passport of the minimal completed architecture

3.1 Name

$$\boxed{TMI\text{-}ToE\ 1.0}$$

Full title:

Interface-Field Architecture for the Unification of Fundamental Interactions

3.2 Status

$$\boxed{\text{A minimal completed architecture, but not a complete physical Theory of Everything.}}$$

That is:

$$\boxed{TMI\text{-}ToE\ 1.0 \neq \text{a complete Theory of Everything}}.$$

3.3 Main object

$$\boxed{\mathcal{M}_I^{ToE} = (\mathcal{M}_U, \mathcal{E}_I, G_I, \mathbb{A}_I, \mathbb{F}_I, \Psi, H, \chi_I, \mathrm{Adm}_I, \mathrm{Struct}_I)}$$

Symbol	Meaning
$\mathcal{M}_U$	spacetime
$\mathcal{E}_I$	universal interface bundle
G_I	universal interface group
$\mathbb{A}_I$	universal interface connection
$\mathbb{F}_I$	universal interface curvature
Ψ	full quantum-field sector
H	Higgs field
χ_I	interface field
Adm_I	set of admissible physical transitions
Struct_I	structured physical result

3.4 Universal bundle

$$\boxed{\pi_I : \mathcal{E}_I \to \mathcal{M}_U}$$

The fiber over a point is:

$$\boxed{\mathcal{F}_{I,x} = \pi_I^{-1}(x)}.$$

Working decomposition:

$$\boxed{\mathcal{F}_{I,x} = \mathcal{F}_G \oplus \mathcal{F}_{SU(3)} \oplus \mathcal{F}_{SU(2)} \oplus \mathcal{F}_{U(1)} \oplus \mathcal{F}_\Psi \oplus \mathcal{F}_H \oplus \mathcal{F}_\chi}$$

3.5 Universal connection

$$\boxed{\mathbb{A}_I \in \Omega^1(\mathcal{M}_U, \mathfrak{g}_I)}.$$

Working form:

$$\boxed{\mathbb{A}_I = \omega^{ab} J_{ab} + e^a P_a + G^A T_A + W^i T_i + BY + A_\chi \Xi}$$

where A_χ is an interface one-form. If χ_I is taken as a scalar field, one may set:

$$\boxed{A_\chi = d\chi_I}.$$

3.6 Universal curvature

$$\boxed{\mathbb{F}_I = d\mathbb{A}_I + \mathbb{A}_I \wedge \mathbb{A}_I}.$$

Projections:

$$\boxed{F_a = \pi_a(\mathbb{F}_I), \qquad a \in \{G, S, W, EM\}}.$$

Main formula:

$$\boxed{\text{The four fundamental interactions are projections of a unified interface curvature.}}$$

3.7 Unified action

$$\boxed{S_{TMI-ToE} = S_{GR} + S_{SM} + S_\chi + S_{mix}}.$$

Expanded form:

$$\boxed{S_{TMI-ToE} = \int_{\mathcal{M}_U} \sqrt{-g}\left[\frac{c^4}{16\pi G}(R - 2\Lambda) + \mathcal{L}_{SM} + \mathcal{L}_\chi + \mathcal{L}_{mix}\right] d^4x}$$

Interface Lagrangian:

$$\boxed{\mathcal{L}_\chi = -\frac{1}{2}g^{\mu\nu}\nabla_\mu\chi_I\nabla_\nu\chi_I - V(\chi_I) - \frac{1}{2}\xi_I R\chi_I^2}$$

Mixing term:

$$\boxed{\mathcal{L}_{mix} = -\frac{1}{4}\sum_{a=1}^{3}\zeta_a\frac{\chi_I^2}{M_I^2}\,\mathrm{Tr}(F_{a\mu\nu}F_a^{\mu\nu}) + \eta_H\chi_I^2 H^\dagger H + \sum_f y_{I,f}\frac{\chi_I}{M_I}\bar{\Psi}_{L,f}H\Psi_{R,f} + h.c.}$$

3.8 Main chain

$$\boxed{\mathcal{E}_I \Rightarrow G_I \Rightarrow \mathbb{A}_I \Rightarrow \mathbb{F}_I \Rightarrow \{F_G, F_S, F_W, F_{EM}\} \Rightarrow \mathrm{Adm}_I \Rightarrow \mathcal{H}_I \Rightarrow \Psi \Rightarrow \mathrm{Dist}_{poss} \Rightarrow \mathrm{Struct}_I}$$

Meaning:

1. the unified interface structure defines admissible interactions;
2. interactions define physical transitions;
3. the quantum state defines the distribution of possibilities;
4. the interface transforms possibilities into structure.

4 Testable consequences

4.1 Consequence 1: corrections to gauge couplings

From the gauge mixing term

$$-\frac{1}{4}\zeta_a\frac{\chi_I^2}{M_I^2}\,\mathrm{Tr}(F_{a\mu\nu}F_a^{\mu\nu})$$

one obtains the effective correction:

$$\boxed{\frac{1}{g_{a,obs}^2(E)} = \frac{1}{g_{a,SM}^2(E)} + \zeta_a \frac{\chi_I^2(E)}{M_I^2}}$$

and therefore:

$$\boxed{\Delta_a(E) = \frac{1}{g_{a,obs}^2(E)} - \frac{1}{g_{a,SM}^2(E)} = \zeta_a \frac{\chi_I^2(E)}{M_I^2}}$$

Null hypothesis:

$$\boxed{H_0 : \zeta_a = 0}.$$

TMI-ToE hypothesis:

$$\boxed{H_{TMI-ToE} : \exists a,\ \zeta_a \neq 0}.$$

4.2 Consequence 2: corrections to the Higgs sector

Through $V(H, \chi_I)$ one obtains:

$$\boxed{\mu_{H,eff}^2 = \mu_H^2 - \eta_H \chi_I^2}$$

and the possible test line:

$$\boxed{\Delta v_H = v_H^{obs} - v_H^{SM} = F(\eta_H, \chi_I)}.$$

4.3 Consequence 3: relation to interface decoherence

If:

$$\Gamma_I = \lambda_I \|\chi_I\|^2$$

and:

$$\Delta_a(E) = \zeta_a \frac{\chi_I^2(E)}{M_I^2},$$

then:

$$\boxed{\Delta_a(E) = \frac{\zeta_a}{\lambda_I M_I^2} \Gamma_I(E)}.$$

This is a cross-module prediction: the same interface intensity may appear both in decoherence and in gauge-coupling corrections.

5 Criteria of support and weakening

5.1 Support criteria

The module receives support if:
1. GR, SM, and QM are correctly recovered;
2. the mixing terms preserve gauge symmetries;
3. no uncancelled anomalies appear;
4. the effects are suppressed by M_I and do not contradict data;
5. observed corrections share a common dependence on χ_I^2.

5.2 Weakening criteria

TMI-ToE weakens if:
1. GR or SM is not recovered;
2. $\mathcal{L}_{mix}$ violates gauge invariance;
3. uncancelled anomalies appear;
4. effective corrections contradict experimental constraints;
5. no distinguishable testable consequence exists.

6 Final fixation

$$\boxed{\text{TMI-ToE 1.0 is closed as a minimal consistent architecture.}}$$

But:

$$\boxed{\text{TMI-ToE 1.0 is not a completed physical Theory of Everything.}}$$

Proper formulation:

> TMI-ToE-1.0 is a completed minimal architectural module of unification in which the four interactions are represented as projections of a unified interface curvature, while the interface field χ_I connects geometry, gauge fields, quantum state, dark sector, black holes, and the process of structuring.

Shortest final formula:

$$\boxed{\mathbb{F}_I = d\mathbb{A}_I + \mathbb{A}_I \wedge \mathbb{A}_I}$$

$$\boxed{F_a = \pi_a(\mathbb{F}_I), \qquad a \in \{G, S, W, EM\}}$$

$$\boxed{S_{TMI-ToE} = S_{GR} + S_{SM} + S_\chi + S_{mix}}$$

$$\boxed{\chi_I \Rightarrow \{\Gamma_I,\ DarkSector,\ \Delta_a(E),\ BH\text{-}interface\}}$$